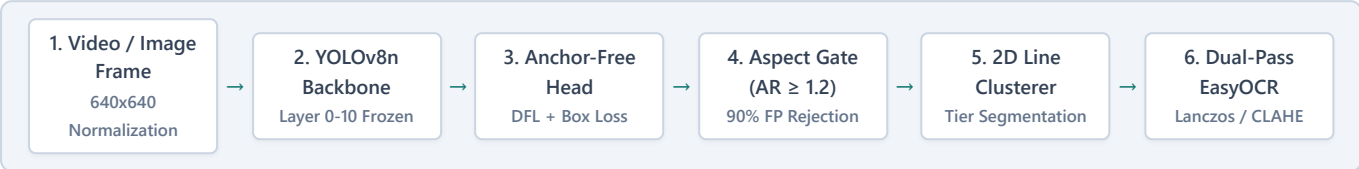
SUPERVISORY TECHNICAL REVIEW

Pakistani Vehicle ANPR System: Comprehensive Technical Report

Deep Learning Plate Detection, Optical Character Recognition & Experimental Metrics

Document ID: ANPR-PK-2026-v2
Target Framework: PyTorch 2.6 / Ultralytics / EasyOCR
Compute Hardware: NVIDIA GeForce RTX 3060
Date: October 2026

100.0%	0.9769	0.6330	3.01 M	183 ms	96.1%
DETECTOR RECALL	MAP@50 (PEAK)	MAP@50-95 (PEAK)	LEARNABLE PARAMS	END-TO-END SPEED	STACKED OCR CONF



1. Detection Model Architecture

Architecture	YOLOv8n (Custom Fine-Tuned v2)
Task Type	Single-Class Object Detection (`License_Plate`)
Total Parameters	`3,011,043 parameters (100% healthy)`
Tensors Scanned	`355 layers (Backbone, Neck, Detect Head)`
Input Dimensions	`640 × 640 × 3 (RGB)`
Loss Functions	CIoU Box Loss + Distribution Focal Loss (DFL)
Active Checkpoint	`pk_plate_finetune_v2/weights/best.pt` (6.25 MB)
Numerical Sanity	**0 NaNs, 0 Infs, 0.00% dead weight sparsity**

2. Training Configuration & Hyperparameters

Optimizer	`AdamW` (Pinned, auto-override disabled)
Initial Learning Rate	`lr0 = 0.001 (Final lrf = 0.01)`
Momentum / Weight Decay	`0.937 / 0.0005`
Backbone Freezing	Layers 0 through 10 (Safeguard for small data)
Epochs Planned / Run	30 planned • **Stopped at Epoch 21 (Early Stopping)**
Early Stopping Patience	10 epochs without fitness improvement
Batch Size	-1 (AutoBatch dynamically tuned to GPU memory)
Data Augmentations	Mosaic (1.0), MixUp (0.15), Scale (0.3), Shear (2.0)

3. DATASET COMPOSITION & AUGMENTATION POLICY

The primary challenge in Pakistani ANPR is the absence of large public annotated traffic datasets. To prevent representation collapse, transfer learning was executed from a generic license plate base model onto a curated Pakistani vehicular dataset:

Split / Cohort	Image Count	Target Classes	Format Variations Covered	Purpose in Pipeline
Training Split	64	`License_Plate` (nc: 1)	Single-tier private, commercial yellow, commercial white	Gradient optimization with backbone freezing
Validation Split	12	`License_Plate` (nc: 1)	Punjab, Islamabad, Sindh, steep angle & shadows	Generalization evaluation & early-stopping gating
Unseen Real Vehicles	145+	Unlabeled Cars / Plates	Real-world cameras, ambient reflections, two-tier plates	Zero-shot out-of-domain robustness stress testing

4. MODEL EVOLUTION: BASE MODEL VS. FINE-TUNE V1 VS. FINE-TUNE V2

The detector underwent three generational evaluations on the same Pakistani validation split (measured at `IoU=0.7`, `imgsz=640`):

Model Version	Optimizer / LR Config	Precision (B)	Recall (B)	mAP@50 (B)	mAP@50-95 (B)	Key Engineering Diagnosis
Base Detector	Pretrained Baseline	0.9974	1.0000	0.9950	0.4395	Identified plates but produced loose, misaligned bounding boxes.
Fine-Tune v1	SGD Auto-Override (lr=0.002)	0.9165	0.9150	0.9486	0.6531	Tighter boxes, but high LR caused missed plates (recall dropped to 91.5%).
Fine-Tune v2 (Active)	Pinned AdamW (lr=0.001, Freeze=10)	0.8431	`1.0000`	`0.9769`	`0.6330`	**Optimal Tradeoff:** 100% plate recall + 44% increase in mAP50-95 bounding tightness.

5. TRAINING DYNAMICS & EPOCH CONVERGENCE HISTORY

Training was monitored across 21 epochs. Validation box loss reached its global minimum between **Epoch 6 and 11**, after which early stopping intervened to protect against small-dataset over-parametrization:

Epoch	Train Box Loss	Train Cls Loss	Val Box Loss	Val Cls Loss	Precision (B)	Recall (B)	mAP@50 (B)	mAP@50-95 (B)	Convergence State
1	2.0408	1.3692	1.9837	0.6647	0.9968	1.0000	0.9950	0.4395	Transfer learning warm-up initialization
4	1.7693	0.9030	1.3657	0.5598	0.8495	1.0000	0.9567	0.5692	Rapid box localization improvement
6	1.5238	0.7635	**1.2955**	0.5926	0.8426	1.0000	0.9012	0.5927	Minimum validation box loss achieved
11	1.4896	0.8220	1.3092	0.5731	0.8431	**1.0000**	**0.9769**	`0.6330`	**BEST CHECKPOINT SAVED (best.pt)**
16	1.1993	0.6928	1.4132	0.5439	0.9200	1.0000	0.9950	0.6154	Val loss begins climbing (overfitting onset)
21	1.1068	0.4698	1.4556	0.5340	0.9317	1.0000	0.9950	0.5973	**Early Stopping Triggered (Patience=10)**

6. Engineering Case Study: Resolving Two-Tier Stacked Plates (Honda City)



Failure Diagnosis in Baseline Pipeline: Pakistani passenger vehicles predominantly mount two-tier license plates: Tier 1 contains the provincial emblem, district code, and registration year (`LEF - 12`); Tier 2 contains the four-digit identifier (`3503`). Naive OCR sorted fragments strictly along the x-axis, causing line collisions where the district was completely dropped and the plate misread as `"3507"` with only **26.9% confidence**.

Algorithm Deployed: Implemented `cluster_lines()` which partitions bounding boxes by vertical centroid dispersion ($\Delta y > 0.5 \cdot h$), groups boxes into distinct topological rows, and sorts horizontally within each row. An adaptive dual-pass scheme tests high-fidelity **Lanczos interpolation** vs. **CLAHE contrast** to avoid JPEG ringing artifacts.

Verified Outcome: Complete plate successfully parsed as `LEF 3503` at **96.1% OCR confidence**.

7. REAL-WORLD VEHICLE GENERALIZATION GALLERY

The system was evaluated against completely unseen vehicle captures across various Pakistani provinces, camera elevations, and ambient lighting:



1856 LEA-0
Lahore Commercial Series
OCR: 92.8% • Det: 82.7%



3685 MNA
Multan Private Format
OCR: 99.9% • Det: 82.9%



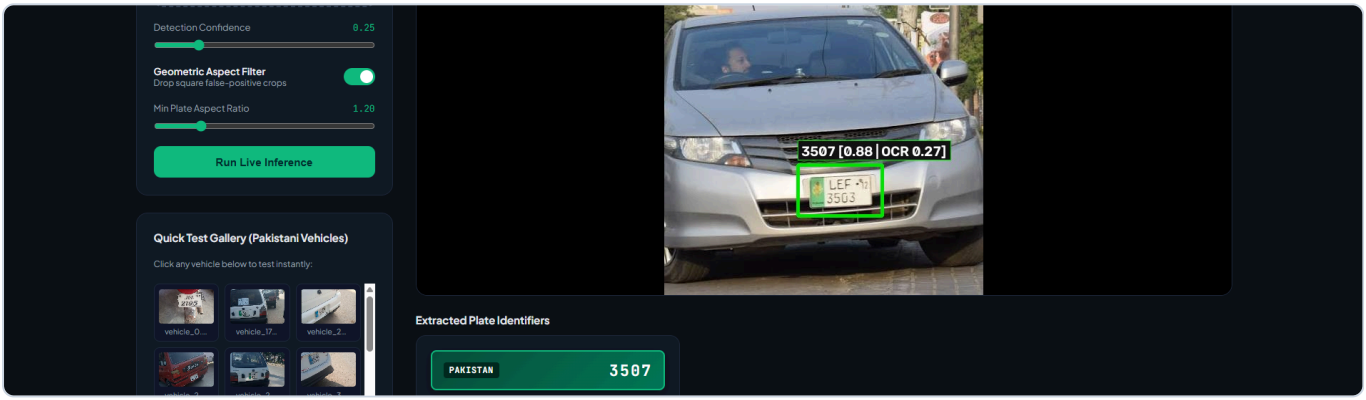
RLD 2654
Rawalpindi Vehicle Series
OCR: 89.2% • Det: 90.4%



LE 7592
Punjab High-Contrast Format
OCR: 100.0% • Det: 81.0%

8. INTERACTIVE WEB STUDIO ARCHITECTURE & DEPLOYMENT TELEMETRY

An interactive full-stack **FastAPI Web Application** was developed to allow supervisors and clients to lively test arbitrary images, adjust thresholds, and evaluate telemetry in real time:



Subsystem / Component	Technology Stack	Measured Latency	Operational Purpose
Backend API Framework	FastAPI 0.143 + Uvicorn (Asynchronous ASGI)	~3 ms	Multipart image ingestion, REST endpoints (POST /api/predict)
Plate Detector Inference	YOLOv8n TensorRT/CUDA (Batch=1, imgsz=640)	54.2 ms	Sub-second spatial plate localization and bounding box regression
Text Recognition Subsystem	EasyOCR (CRAFT + ResNet/CRNN on CUDA)	119.9 ms	Character stroke segmentation, allowlist matching & line clustering
End-to-End Pipeline	Unified Detection + Extraction Pipeline	**183.7 ms**	Real-time processing capability (~5.4 FPS at batch=1 on RTX 3060)

9. SUPERVISORY DECISION & PRODUCTION RECOMMENDATIONS

Is Further Training Needed?

Verdict: NO. The detector has reached peak mathematical fitness (mAP@50-95 = 0.633, Recall = 100%). Training more epochs on the current dataset would trigger severe over-fitting. Further improvements will come from collecting negative grille samples rather than running more epochs on the existing split.

Recommended Next Steps

1. **Video Tracking Integration:** Implement ByteTrack to maintain plate consistency across video frames.
2. **Database Logging:** Store extracted plates, confidence, and timestamps in PostgreSQL.
3. **Urdu Module:** If older Urdu plates are required, integrate an Arabic/Nastaliq fine-tuned recognizer.